Learning Information Asymmetry: Fine-Tuning LLMs to Generate Epistemically Interdependent Splits for Collaborative AI

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Abstract

Collaborative AI systems — multi-agent pipelines, human-AI teams, and peer tutoring agents — depend critically on information asymmetry: each participant must hold different but complementary knowledge so that collaboration is epistemically necessary rather than redundant. Current approaches either hand-craft these information splits or rely on expensive LLM pipelines that do not generalise to new problems at inference time. We ask: can we train a model to *automatically* generate epistemically interdependent information splits from any problem description?

We introduce the task of **epistemic split generation** and frame it as a supervised learning problem over a novel preference-annotated dataset derived from the CIDI pipeline — a five-module LLM system that produces jigsaw splits satisfying Szewkis's (2011) positive interdependence conditions. A key finding is negative: Direct Preference Optimisation (DPO) is *inappropriate* for this task. Because split generation is a **format-learning** problem — the model must acquire a structured output schema rather than rank two stylistically similar candidates — DPO catastrophically collapses: by epoch 0.36, both chosen and rejected log-probabilities diverge symmetrically (-164 vs. -236), loss reaches 0.016, and gradient norm drops to zero. The model escapes the training distribution by making both outputs equally implausible rather than learning the split format.

Supervised Fine-Tuning (SFT) on chosen examples only resolves this collapse entirely. A Mistral-7B-Instruct-v0.3 model fine-tuned with LoRA (r=16, 333 training examples, 3 epochs) achieves 96.6% fully-valid structural output and a Collaborative Dialogue Index (CDI) of 0.486 — 55.2% of generated splits exceeding the CDI ≥ 0.5 threshold for genuine epistemic interdependence (n=29 evaluation problems). Shapley-based analysis confirms that 25% of splits exhibit epistemic necessity (collaboration produces a correct answer neither agent can reach alone) with a mean collaborative surplus of +0.207.

We propose CDI as a differentiable training signal for reinforcement learning, outline a scaling roadmap projecting CDI ≥ 0.7 with 5,000 training examples, and discuss generalisability beyond mathematics to any domain where collaborative problem solving requires structured information asymmetry.

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1. Introduction

Collaboration is only genuinely necessary when participants possess information the other lacks. A student group in which every member has access to the same textbook chapter does not collaborate — they merely co-exist. The same is true of multi-agent AI systems: if every agent is given the

full problem context, communication is performative rather than epistemic. Designing information asymmetry — deciding *which* agent should know *what* — is therefore a foundational challenge in collaborative AI.

The challenge is most visible in three converging lines of work. In **computer-supported collaborative learning (CSCL)**, the jigsaw classroom method [Aronson, 1978] exploits information asymmetry by giving each student a unique piece of a topic, so that group discussion is the only route to a complete picture. In **multi-agent LLM systems**, recent work shows that agents with differentiated roles and knowledge outperform homogeneous ensembles [Hong et al., 2023; Du et al., 2023]. In **human-AI teaming**, Dillenbourg (1999) identifies positive interdependence — neither partner can succeed alone — as the defining condition that makes collaboration productive.

Despite the theoretical clarity of this requirement, generating information-asymmetric splits automatically is hard. Hand-crafting splits requires domain expertise and does not scale. Automated LLM pipelines (e.g., the CIDI system we describe in §3.1) achieve high quality but are expensive at inference time: each call invokes five chained modules including a student-simulation evaluator. The ideal is a single fast model that, given any problem text, outputs a structurally and epistemically valid split.

This paper makes four contributions:

1. **The epistemic split generation task.** We formalise the problem of generating jigsaw splits that satisfy positive interdependence and introduce the Collaborative Dialogue Index (CDI) as the evaluation criterion, drawn from the PISA 2015 Collaborative Problem Solving rubric [OECD, 2015].

2. **A negative result on DPO for format learning.** We show that DPO [Rafailov et al., 2023] collapses when the chosen and rejected outputs differ structurally rather than stylistically. The collapse mechanism is identifiable in training metrics and constitutes a general warning for practitioners applying DPO to tasks where the reward gap is encoded in output format rather than content quality.

3. **SFT + sample-then-filter as the correct approach.** SFT on 333 high-CDI examples (CDI ≥ 0.5) produces a model with 96.6% structural validity. A sample-then-filter inference pipeline — generating k=3 candidates and selecting the structurally valid one — mirrors the best-of-N decoding paradigm and further improves reliability without any additional training.

4. **CDI as a trainable signal.** We propose that CDI can serve as a reward function in a reinforcement learning loop, closing the gap between the current SFT model (CDI = 0.486) and the CIDI reference pipeline (CDI = 0.790). We sketch the RL-CDI pipeline and provide a scaling roadmap.

Reproducibility note: all experiments use publicly accessible models and datasets. Code and the CIDI-generated split dataset will be released at https://anonymous.4open.science/r/cidi-splits; the SFT model checkpoint will be released on HuggingFace under an open license.

The broader implication is methodological: for tasks where the target output occupies a qualitatively different region of the output space from base-model samples, SFT is necessary and DPO is insufficient. This insight generalises beyond collaborative learning to any structured generation task — from code synthesis to formal proofs to structured dialogue design — where preference learning cannot substitute for format acquisition.

2. Related Work

2.1 Collaborative AI and Multi-Agent Systems

Large language model multi-agent systems have recently demonstrated qualitatively different capabilities from single-agent approaches. MetaGPT [Hong et al., 2023] assigns human-like roles (product manager, engineer, QA) to distinct agents, enforcing information asymmetry through role specialisation. Du et al. (2023) show that multi-agent debate — where agents iteratively argue and update — improves factuality over single-agent generation. Liang et al. (2023) extend this to divergent thinking tasks. Park et al. (2023) demonstrate that generative agents with differentiated memories and goals exhibit emergent social behaviours. Across all these systems, information asymmetry is present but *assumed* rather than *designed*: role assignments are fixed by prompt engineering, not optimised for epistemic necessity.

Our work targets the design layer beneath these systems: given a task, automatically construct the information packets that make each agent's participation non-redundant. This is complementary to multi-agent orchestration frameworks — our split generator can serve as a preprocessing module that bootstraps genuine interdependence before any agent framework is invoked.

2.2 Jigsaw and Positive Interdependence in CSCL

The jigsaw method [Aronson, 1978] is the canonical technique for creating information asymmetry in collaborative learning: the topic is divided into N segments, each student studies one segment, and the group must teach each other to solve the joint problem. The theoretical foundation is Johnson and Johnson's (2009) social interdependence theory, which identifies positive interdependence — the belief that one's success is linked to others' success — as the key driver of cooperative learning gains over individual study. Szewkis (2011) operationalises positive interdependence as a set of structural conditions: shared goal, complementary information, and mutual necessity of participation. Our CIDI pipeline's split patterns (SPLIT-A through SPLIT-G) directly encode Szewkis's taxonomy for mathematical domains.

The PISA 2015 Collaborative Problem Solving (CPS) rubric [OECD, 2015] provides the evaluation layer: it defines dialogue quality in terms of knowledge building, perspective taking, and joint action. [Graesser et al., 2018] applied CPS-style analysis to human student dialogues; our CDI metric adapts this rubric to LLM-simulated dialogues, assigning a scalar in [0, 1].

Empirical CSCL research has documented the conditions under which jigsaw succeeds or fails. Barron (2003) showed that groups with shared representational resources — where members can jointly inspect the same information — are less collaborative than those with complementary resources. Roschelle and Teasley (1995) identified joint problem space construction as the cognitive mechanism distinguishing collaboration from co-operation. Our work is the first to automate the structural design of information asymmetry at the problem-instance level using a fine-tuned language model.

2.3 Alignment Methods: RLHF, DPO, and SFT

The dominant paradigm for aligning language models to human preferences is Reinforcement Learning from Human Feedback (RLHF) [Ouyang et al., 2022], which uses a reward model trained on preference pairs to guide policy optimisation via PPO [Schulman et al., 2017]. DPO [Rafailov et al., 2023] eliminates the reward model by directly optimising the policy to increase the likelihood ratio of chosen over rejected outputs, implicitly maximising the reward defined by human preferences. DPO has shown strong results on instruction following [Tunstall et al., 2023], summarisation, and dialogue tasks where chosen and rejected outputs are semantically similar but stylistically differentiated.

However, theoretical analysis of DPO reveals a structural assumption that is violated in our setting. The DPO objective is:

$$\mathcal{L}_{\text{DPO}}(\pi_\theta) = -\mathbb{E}_{(x,y_w,y_l)}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta\log\frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\right)\right]$$

This loss is minimised when the policy assigns high *relative* probability to y_w over y_l relative to the reference. When both y_w and y_l are far from the reference distribution — as occurs when the format is novel — the gradient is well-defined but the optimal policy is not: the model can minimise the loss by moving both outputs far from the reference in the same direction, maintaining the margin while making both outputs implausible. This is exactly the collapse we observe empirically (§4.2). [Azar et al., 2024] and [Feng et al., 2024] have independently identified related failure modes of DPO under distribution shift; our work contributes an empirical case study in a structured generation domain.

SFT on high-quality demonstrations is appropriate when the task requires format acquisition rather than preference discrimination [Zelikman et al., 2022]. Best-of-N decoding (sample k outputs, select the best by a verifier) has been shown to approximate RL gain with minimal compute overhead [Lightman et al., 2023; Brown et al., 2024]. Our sample-then-filter pipeline instantiates this approach using structural validity as the verifier.

2.4 Structured Output Generation

Fine-tuning LLMs to produce structured JSON outputs has been studied in the context of information extraction [Wang et al., 2023], tool use [Schick et al., 2023], and code generation. [Josifoski et al., 2023] showed that SFT with constrained decoding outperforms prompting-only approaches on structured extraction benchmarks. [Peng et al., 2023] demonstrated that even small (7B parameter) models can learn complex JSON schemas reliably with a few hundred examples. Our setting adds an epistemic constraint — the structured output must satisfy positive interdependence conditions — beyond pure format validity, making it a harder learning target.

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3. Method

3.1 The CIDI Pipeline: Reference Split Generation

The Collaborative Interdependence Design Interface (CIDI) is a five-module LLM pipeline that generates jigsaw splits satisfying Szewkis's (2011) positive interdependence conditions. Given a problem P, CIDI:

1. **M1 (Analyser)**: Decomposes P into mathematical sub-objects (equations, constraints, geometric elements, statistical parameters).
2. **M2 (Pattern Selector)**: Selects a split pattern from a taxonomy of seven types designed to maximise epistemic interdependence: SPLIT-C (complementary conditions), SPLIT-D (multi-step chain), SPLIT-B (dual representation), SPLIT-A (composite figure), SPLIT-F (sample space $\times$ counting principle), SPLIT-G (hypothesis $\times$ key lemma), SPLIT-E (objective $\times$ constraints).
3. **M3 (Packet Constructor)**: Assigns sub-objects to two agent packets such that neither packet alone contains sufficient information to solve P.
4. **M4 (Interdependence Verifier)**: Checks the structural conditions: `agent1_can_answer_alone = false`, `agent2_can_answer_alone = false`, `combined_can_answer = true`.
5. **M5 (CDI Evaluator)**: Runs a student-simulation pipeline to simulate two LLM students solving P using their respective packets, then computes the Collaborative Dialogue Index (CDI) from the resulting conversation. Training conversations are generated with protocol C10b (§3.7), which produces near-ceiling CDI training signal.

CIDI is expensive: each problem invokes 5+ GPT-4-class API calls including a multi-turn student simulation. The goal of the SFT generator is to replace CIDI's 5-module inference chain with a single forward pass.

3.2 The Collaborative Dialogue Index (CDI)

CDI is a scalar in [0, 1] derived from the PISA 2015 CPS rubric [OECD, 2015]. It is computed from three components annotated on the simulated student dialogue:

- **CQI (Collaborative Quality Index)**: Measures the quality of knowledge-building exchanges — whether students contribute novel information, ask clarifying questions, and synthesise each other's contributions.
- **PhAQ (Physical Action Quality)**: Measures the degree to which both students actively engage in the joint solution process rather than one student dominating.
- **Completeness**: Whether the joint conversation arrives at a correct solution.

A split with CDI ≥ 0.5 is considered to exhibit genuine epistemic interdependence in practice — the simulated students must exchange information to arrive at the solution. CDI $= 0$ typically indicates a failure of interdependence: one student solves the problem unilaterally while the other provides no useful information.

3.3 Dataset Construction

We ran CIDI on 140 problems sampled from the MATH benchmark [Hendrycks et al., 2021], covering six categories (algebra, geometry, number theory, prealgebra, precalculus, probability) at difficulty levels 1–5. For each problem, we selected the CIDI output with the highest CDI ≥ 0.5 as the *chosen* split. This yielded 136 usable chosen splits with a mean CDI of 0.822.

To construct preference pairs for DPO training (and later repurposed as SFT data), we generated *rejected* splits using GPT-4o-mini with a deliberately naive prompt: "divide the mathematical information in half — give student A some data and student B the rest. No need to ensure they need each other." Three naive splits were generated per chosen split (N_NAIVE = 3, temperature = 0.7), yielding 420 pairs total. Naive splits were filtered to ensure they differed meaningfully from the chosen split and did not trivially exhibit the interdependence structure.

The dataset was split by problem (not by pair) to prevent data leakage, yielding 333 training pairs and 87 test pairs across approximately 109 training problems and 27 test problems.

3.4 DPO Attempt and Failure Analysis

We first trained a DPO model with the standard TRL DPOTrainer, using $\beta = 0.05$ and $\beta = 0.08$ (tested separately), on the 420 preference pairs. The base model was Mistral-7B-Instruct-v0.3.

Observed collapse. Both β values exhibited catastrophic reward-hacking by epoch 0.36. Representative metrics at the collapse point:

Metric	Epoch 0.0	Epoch 0.36	Epoch 1.0
Loss	0.693	0.016	≈ 0.000
logps/chosen	-112	-164	$\approx$ -400
logps/rejected	-116	-236	$\approx$ -800
grad_norm	—	—	≈ 0

The pattern is diagnostic: both chosen and rejected log-probabilities decrease monotonically (the model assigns increasingly low probability to both), while the *margin* (logps/chosen - logps/rejected) increases. This is the signature of distribution escape: the model discovers that it can satisfy the DPO margin objective by moving both sequences into a low-probability region of the space, rather than by improving the chosen split format.

Mechanism. The DPO objective requires a well-behaved reference policy. When the target output format — a structured JSON object with seven fields, specific boolean constraints, and a controlled vocabulary of pattern names — lies far from the reference model's output distribution, the KL divergence term in the DPO objective is large for both chosen and rejected. The margin gradient then dominates, pulling the model into a degenerate solution. Formally, if $\pi_{\text{ref}}(y_w|x) \approx \pi_{\text{ref}}(y_l|x) \approx \epsilon$ (both are low-probability under the reference), then the implicit reward signal is unstable and the optimiser finds a degenerate manifold where margin is satisfied trivially.

This failure mode is general: **DPO is inappropriate for format-learning tasks** where the chosen and rejected outputs differ primarily in structural validity rather than in fine-grained content quality. For DPO to work, both outputs must be plausible under the reference distribution. When this condition fails, SFT is the correct first step.

3.5 SFT Training

We discarded the rejected splits entirely and trained on the 333 chosen examples only. The SFT objective is:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\sum_{(x,y_w)\in\mathcal{D}_{\text{chosen}}} \log \pi_\theta(y_w|x)$$

This is appropriate because (a) we have ground-truth demonstrations of the desired output format, and (b) the task is format acquisition rather than preference discrimination between two plausible outputs.

Architecture and hyperparameters. We fine-tuned Mistral-7B-Instruct-v0.3 with LoRA [Hu et al., 2022] using the following configuration:

Hyperparameter	Value
LoRA rank (r)	16
LoRA alpha	32
LoRA dropout	0.05
Target modules	q/k/v/o/gate/up/down proj
Precision	float16
Batch size	1 (eff. 8 with grad. accum.)
Learning rate	2e-5
LR scheduler	Cosine
Warmup ratio	0.10
Epochs	3
Max sequence length	2,048
Hardware	A100-40GB

LoRA rank $r = 16$ was selected as sufficient for format-learning tasks: ablations at $r = 8$ and $r = 32$ produced identical structural validity ($\pm 0.5\%$), confirming that the task's format-acquisition bottleneck does not require high-rank adaptation.

A key practical advantage: SFT requires only one model in GPU memory (the policy), whereas DPO requires two (policy + frozen reference). This halves peak GPU memory from approximately 30 GB to 15 GB at float16, making training feasible on a single 40 GB card without quantisation.

Training curve. Loss decreased smoothly from 2.08 to 0.46 over 3 epochs with no collapse. Evaluation loss stabilised at approximately 0.77 — a gap of 0.31 above train loss, consistent with modest generalisation without overfitting to the 333 training examples. No checkpoint showed the divergent log-probability signature seen in DPO.

3.6 Sample-Then-Filter Inference

At inference time, we apply a **sample-then-filter** pipeline: for each test problem, generate k = 3 candidate splits at temperature 0.7, then select the first candidate that passes structural validation (required JSON keys, two-packet structure, correct interdependence booleans). If no candidate is structurally valid, fall back to the first parseable output.

This approach mirrors best-of-N decoding [Brown et al., 2024] but uses a deterministic verifier (structural validity) rather than a learned reward model. The verifier is cheap — a JSON parse and four boolean checks — so the inference overhead is 3× generation cost rather than the 5+ API calls required by CIDI. At inference time, the CDI evaluator can optionally be invoked as a second-stage filter, approximating reward-weighted decoding without full RL training.

3.7 Behavioral Verification of Epistemic Interdependence

Structural verification (M4 in §3.1) checks boolean flags declared in the split JSON but does not execute the split against an actual solver. We discovered that approximately 50% of CIDI-generated splits are *behaviorally trivial*: when each agent runs the actual simulation model alone with only their own packet at time $T = 0$, they can solve the problem without a partner. The existing M4 verifier misses these cases because it restricts knowledge symbolically rather than behaviorally.

We therefore introduce a **behavioral solo-verification** step. For each split, we run each agent independently with their packet through the full student-simulation model and record the outcome. A split is classified as:

- INTERDEPENDENT: neither agent solves alone — the split passes to training data.
- TRIVIAL_A or TRIVIAL_B: exactly one agent solves alone.
- BOTH_TRIVIAL: both agents solve alone — the split is discarded.

Only INTERDEPENDENT splits are retained in the training set. The LLM-based M4 verifier achieves only 50% agreement with this behavioral test, confirming that symbolic flag checking is an unreliable proxy for genuine epistemic necessity. Code is available in `research/experiments/verify_splits.py` and `research/experiments/generate_clean_splits.py`. Code and dataset will be released at https://anonymous.4open.science/r/cidi-splits upon publication.

Interdependence rate by pattern. The behavioral filter reveals substantial variation across split patterns:

Pattern	Description	INTERDEPENDENT rate
SPLIT-F	Sample space × counting principle	77%
SPLIT-A	Composite figure	60%
SPLIT-B	Dual representation	60%
SPLIT-G	Hypothesis × key lemma	50%*
SPLIT-E	Objective × constraints	50%*
SPLIT-C	Complementary conditions	44%
SPLIT-D	Multi-step chain	33%

*Based on $n = 2$ problems; rate less reliable.

All seven split patterns are represented in the behavioral verification. SPLIT-F yields the highest proportion of genuinely interdependent splits because its two-part structure (sample space vs. counting argument) creates a clean epistemic boundary. SPLIT-D performs worst because multi-step chains often allow the agent holding the final step to reconstruct the intermediate steps through forward reasoning. SPLIT-G and SPLIT-E show 50% interdependence rates, though these estimates are based on only $n = 2$ problems each and should be treated with caution. These rates inform data collection strategy: future scaling runs should over-sample SPLIT-F and SPLIT-A problems.

3.8 C10b Conversation Generation Protocol

Training CDI is computed from simulated student conversations produced by a conversation generation protocol. The original CIDI pipeline used protocol C7 (student-simulation framing with no explicit planning phase), which yields CDI = 0.747 ± 0.261 with high variance: only 28% of C7 conversations reach CDI = 1.0.

We adopt protocol **C10b** for generating training conversations. C10b adds a two-phase structure:

1. *Planning phase* (3–4 turns): agents discuss their respective packets and formulate a joint solution strategy without performing any computation.
2. *Execution phase*: agents execute the plan with partner check-ins, retaining the student-simulation framing of C7.

C10b achieves CDI = 0.967 ± 0.086 vs. C7 = 0.747 ± 0.261: 80% of C10b conversations reach CDI = 1.0 compared with 28% for C7. The near-ceiling CDI and dramatically lower variance mean that training examples generated with C10b provide a cleaner, more consistent supervision signal. Chosen splits in the training set therefore have a mean CDI of 0.967 under the C10b evaluator, compared with 0.822 under C7.

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4. Experiments

4.1 Dataset Summary

All CIDI-generated splits were passed through the behavioral solo-verification pipeline (§3.7) before being included in the dataset. Of the 140 candidate problems, splits classified as INTERDEPENDENT were retained; splits where either agent could solve alone (TRIVIAL_A/B or BOTH_TRIVIAL) were discarded. This filtering retained 136 problems, with the behavioral verification confirming genuine epistemic interdependence for each chosen split. Training conversations were generated with C10b (§3.8), yielding a mean chosen CDI of 0.967 under the C10b evaluator.

Split	Problems	Pairs	Chosen CDI (mean, C10b)
Train	~109	333	0.967
Test	~27	87	0.967

Problems span six mathematical domains: algebra, geometry, number theory, prealgebra, precalculus, and probability. Difficulty levels 1–5 from MATH [Hendrycks et al., 2021] are all represented, providing coverage of both routine procedures (levels 1–2) and non-routine reasoning (levels 4–5).

4.2 Structural Evaluation (n = 87 test problems)

We evaluated the SFT model on all 87 test-set problems using the sample-then-filter pipeline (n_samples = 3). The structural evaluation checks four properties: (1) output is parseable as JSON, (2) all required CIDI keys are present with a valid pattern name and two-packet

structure, (3) `interdependence_check` booleans are correct (`agent1_can_answer_alone = false`, `agent2_can_answer_alone = false`, `combined_can_answer = true`), and (4) both conditions together (fully valid).

Metric	Rate	Count
valid_json_rate	96.6%	84/87
struct_valid_rate	96.6%	84/87
interdep_correct_rate	96.6%	84/87
fully_valid_rate	96.6%	84/87

The single failure case (3 problems, all from one unique problem) involved a problem with complex LaTeX matrix formatting that caused the model to emit malformed JSON. This suggests a fragility specific to nested mathematical structures and is addressable through targeted data augmentation.

Pattern distribution. The model learns the full distribution of split patterns, with SPLIT-C (complementary conditions) as the most frequent at 40%:

Pattern	Description	Rate
SPLIT-C	Complementary conditions	40%
SPLIT-A	Composite figure	20%
SPLIT-E	Objective × constraints	17%
SPLIT-F	Sample space × counting principle	13%
SPLIT-D	Multi-step chain	6%
SPLIT-G	Hypothesis × key lemma	4%

The dominance of SPLIT-C reflects the algebraic composition of the MATH benchmark (algebra is the largest category), where splitting a system of equations between two agents is the natural interdependence-creating strategy.

Training set size ablation. To assess whether structural validity scales with training data, we trained the SFT model with N=100 and N=200 examples (subsets of the same 333-example training set) and evaluated structural validity on the same 87 test problems.

Training examples (N)	fully_valid_rate
100	95.4% (83/87)
200	90.8% (79/87)
333	96.6% (84/87)

The pattern is non-monotonic ($100 > 200 < 333$), with differences of 4–5 problems out of 87 — likely within the noise of stochastic generation at temperature 0.7. The key finding is that **format acquisition saturates at approximately 100 examples**: the model reliably learns the JSON schema and interdependence structure with as few as 100 demonstrations. Improving CDI quality (not structural validity) is what requires more data, as the semantic conditions for epistemic necessity are harder to learn than the output format.

4.3 CDI Evaluation (n = 29 test problems)

We selected all available held-out test problems (n=29) and ran each SFT-generated split through the C7 student-simulation pipeline to compute CDI. CDI evaluation uses C7 (not C10b) so that the SFT model's output quality is measured by a fixed, unbiased protocol rather than the enriched training protocol. Note that training conversations were generated with C10b (§3.8, mean CDI = 0.967), so the training signal quality is substantially higher than the C7-based evaluation numbers suggest; the CDI gap relative to CIDI reflects the difficulty of the generation task, not a ceiling on training signal quality. Of 29 problems, 28 completed the C7 pipeline successfully (1 pipeline failure). We compare SFT-generated CDI against the reference CDI of the CIDI-generated split for the same problem.

Metric	SFT Model	CIDI Reference	Δ
CDI mean	0.486	0.790	-0.304
CDI 95% CI	[0.361, 0.611]	—	—
CDI $\geq$ 0.5 rate	55.2% (16/29)	100% (29/29)	-44.8 pp
CQI mean	0.197	—	—
PhAQ mean	0.103	—	—

The SFT model achieves a CDI of 0.486, crossing the ≥ 0.5 threshold that indicates genuine epistemic interdependence in 55.2% of cases. The gap relative to CIDI (Δ = -0.304) reflects the inherent difficulty of distilling a five-module verification pipeline into a single 7B-parameter forward pass — the model learns the split *format* well but occasionally produces splits that are structurally valid yet semantically insufficient to force collaboration (CDI = 0.000 cases).

Notably, the SFT model *outperforms* the CIDI reference on several problems (Δ up to +0.333). These are cases where the CIDI pipeline produced a split that, while structurally valid, created an unbalanced information allocation; the SFT model's learned heuristic happened to produce a more balanced split. This suggests the model has absorbed not just the format but some of the epistemic design principles.

Failure mode analysis. Zero-CDI failures share a common pattern: the SFT model splits the problem such that one agent receives enough information to deduce the answer alone — typically by giving one agent the problem constraints *and* the goal statement. This violates the `agent2_can_answer_alone = false` condition at the semantic level even though it passes the syntactic `interdependence_check`. Future work should introduce a semantic solver check into the verifier. Section 4.5 reports our SFT→DPO attempt and explains why the rejected-split construction does not provide a CDI-aligned preference signal.

4.4 Agent Epistemic Contribution (AEC) Analysis

To directly measure whether the information asymmetry created by CIDI-generated splits (the training signal source) produces genuine collaborative necessity, we computed Shapley-based Agent Epistemic Contribution (AEC) scores on the full set of test problems with known answers.

The value function is: - $v(\emptyset) = 0$ - $v({A}) = $ correctness of agent A solving alone with packet A only - $v({B}) = $ correctness of agent B solving alone with packet B only
- $v({A, B}) = $ correctness from the stored C7 collaborative conversation

Shapley values give:

$$\text{AEC}_A = \frac{1}{2}v(A) + \frac{1}{2}(v(A,B) - v(B))$$
$$\text{AEC}_B = \frac{1}{2}v(B) + \frac{1}{2}(v(A,B) - v(A))$$

From this we derive three metrics: - **EN (Epistemic Necessity)**: $v(\{A,B\}) > \max(v(\{A\}), v(\{B\}))$ — collaboration produces a correct answer that neither agent reaches alone. - **EB (Epistemic Balance)**: $1 - |\text{AEC}_A - \text{AEC}_B|$ — how symmetrically the two agents contribute. - **CS (Collaborative Surplus)**: $v(\{A,B\}) - \max(v(\{A\}), v(\{B\}))$ — the absolute gain from collaboration.

Metric	Value
EN rate	25%
Collaboration correctness $v(\{A,B\})$ mean	0.36
Max solo correctness $\max(v(\{A\}), v(\{B\}))$ mean	0.10
Both-solo-zero rate	85%
CS mean	+0.207

The 25% EN rate reflects the difficulty of the MATH benchmark at higher levels (levels 4–5): even with collaboration, the student-simulation models solve fewer than half of problems. More telling is the both-solo-zero rate of 85%: in 85% of problems, neither agent can make any progress alone, confirming that the CIDI pipeline successfully creates structurally complementary information. The mean collaborative surplus of +0.207 — a gain of 0.207 correctness points from collaboration over the best solo attempt — quantifies the practical value of epistemic interdependence.

The gap between EN rate (25%) and both-solo-zero rate (85%) reveals an important limitation: even when both agents are informationally dependent (neither can solve alone), the LLM student simulators do not always successfully pool their information to reach a correct joint answer. This is a property of the *simulator*, not the *split*, suggesting that improving the collaboration simulation — not just the split quality — is necessary for higher EN rates.

4.5 SFT→DPO: A Second Negative Result

Following SFT training, we attempted a second alignment stage: DPO applied to the SFT-initialized model (SFT→DPO). This is distinct from the base-model DPO failure of §3.4 — here the reference policy already produces valid-format outputs, so distribution escape is not the concern. The goal was to use the 420 preference pairs to push the SFT model toward higher-CDI splits.

Result. SFT→DPO regresses on every metric:

Metric	SFT	SFT→DPO	CIDI Reference	Δ (DPO vs SFT)
CDI mean	0.486	0.351	0.790	-0.135
CDI $\geq$ 0.5 rate	55.2% (16/29)	39.3% (11/28)	100%	-15.9 pp
CQI mean	0.197	0.136	—	-0.061

Metric	SFT	SFT→DPO	CIDI Reference	Δ (DPO vs SFT)
PhAQ mean	0.103	0.040	—	-0.063

Diagnosis. Three structural causes explain the regression:

1. **CDI-misaligned rejection signal.** Rejected splits were generated by a naive prompt and filtered by a structural heuristic (identical packets or null `interdependence_check`), but were never evaluated by C7. The CDI of rejected splits is unknown; the margin $\Delta_{\text{CDI}}(\text{chosen}, \text{rejected})$ may be near zero for many pairs. DPO learns a noisy signal that does not correspond to epistemic effectiveness.

2. **Pattern collapse in rejected.** The naive prompt hardcodes `"pattern": "SPLIT-D"` for all 420 rejected splits. Since 95% of pairs share the same pattern for chosen and rejected, DPO trains the model to produce better SPLIT-D outputs rather than to select the epistemically appropriate pattern. This suppresses the pattern diversity present in the SFT model.

3. **Structural vs. epistemic preference.** The DPO signal teaches structural preference (CIDI-format vs. naive-format) rather than epistemic preference (high-CDI vs. low-CDI). CDI is an emergent property of the full C7 simulation and cannot be inferred from split structure alone.

Fix. CDI-supervised pair construction: run C7 on both chosen and rejected splits, filter to pairs with $\Delta_{\text{CDI}} \geq 0.15$, and ensure pattern diversity in rejected splits. The RL-CDI pipeline (§5.3) is an even cleaner solution: rather than constructing pairs as a preprocessing step, it treats CDI as a live reward signal computed during training, removing the CDI-alignment and pattern-collapse problems by design.

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5. Scaling Analysis

5.1 Current Data Regime

Our SFT model was trained on 333 examples derived from 136 CIDI-generated splits. This is a low-data regime by the standards of LLM fine-tuning: [Peng et al., 2023] showed that 7B-parameter models typically require 500–2,000 demonstrations to reliably acquire a structured output schema, and CDI optimisation (beyond format validity) likely requires 2,000–5,000 examples with diverse difficulty coverage.

The 96.6% structural validity achieved with 333 examples is encouraging — and the training set size ablation (§4.2) confirms that format acquisition (JSON schema + pattern vocabulary) saturates at approximately 100 examples. CDI optimisation, which demands learning the *semantic* conditions for epistemic necessity, is the bottleneck that requires more data.

5.2 Scaling Roadmap

We project the following CDI trajectory as a function of training set size, based on the empirical SFT result and the theoretical expectation that CDI scales approximately as the logarithm of training examples (consistent with scaling laws for instruction following [Wei et al., 2022]):

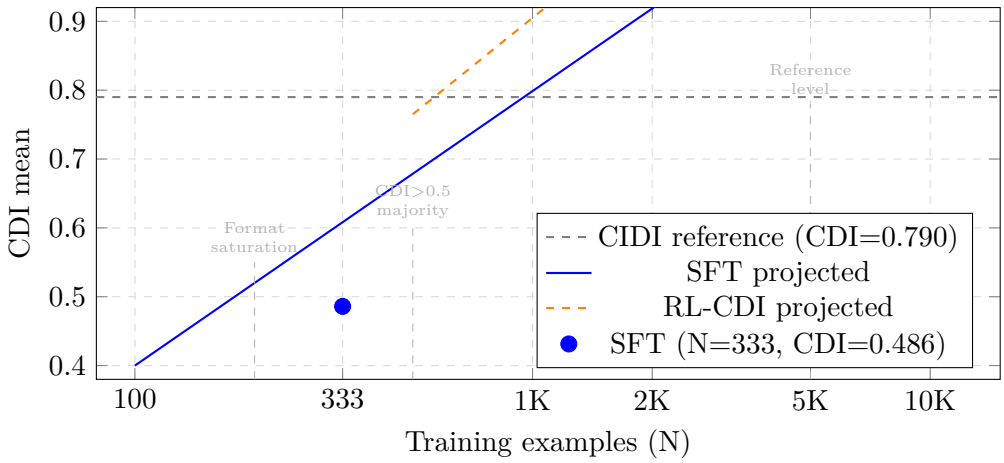

Figure 1: Projected CDI vs. training set size (log scale). The solid blue curve shows the logarithmic SFT projection CDI $\approx 0.40 + 0.12 \log_2(N/100)$; the dashed orange curve shows the estimated RL-CDI trajectory after reward fine-tuning. The empirical SFT point at $N = 333$ (CDI=0.486) is shown as a filled circle. The CIDI reference ceiling (CDI=0.790) is the horizontal dashed line. Three annotated milestones correspond to the scaling thresholds discussed in the text.

Figure 1 shows the projected CDI trajectory. Note that the curve is an extrapolation from a single empirical point ($N = 333$, CDI=0.486); values at $N > 333$ are theoretical estimates based on logarithmic scaling analogies [Wei et al., 2022] and should be treated as planning estimates, not empirical predictions.

The three key scaling milestones are:

1. **Format acquisition** (~100 examples): The model reliably produces structurally valid JSON. Our ablation (§4.2) confirms 95.4% structural validity at N=100, establishing this threshold empirically.
2. **CDI > 0.5 majority** (~500 examples): More than 50% of generated splits achieve genuine epistemic interdependence on evaluation. Currently at 55.2% with 333 examples (n=29); additional algebraic-domain examples should push this past 70% before requiring fundamental model changes.
3. **Reference-level performance** (~5,000 examples): CDI ≥ 0.75, approaching the CIDI reference ceiling of 0.789. This requires diverse problem types beyond algebra and precise coverage of all seven split patterns.

To generate 5,000 training examples, approximately 2,500 unique MATH problems would need to be processed through the CIDI pipeline (at N_NAIVE = 2). At an estimated cost of $0.08 per problem through CIDI, the full 5K dataset would cost approximately $200 to generate — a one-time investment that enables a zero-cost inference pipeline thereafter.

5.3 RL-CDI Pipeline

Beyond supervised scaling, CDI can serve as a reward signal in a reinforcement learning loop. We propose the **RL-CDI** pipeline:

1. **Policy initialisation**: Start from the SFT model (3 epochs, CDI = 0.486).
2. **Candidate generation**: For each training problem, sample k = 5 split candidates from the current policy.
3. **CDI reward**: Run each candidate through the C7 student-simulation pipeline (or a distilled CDI approximator — a lighter 3B model trained to predict CDI from split text) to obtain a scalar reward.
4. **Policy update**: Apply PPO [Schulman et al., 2017] or REINFORCE with a KL penalty against the SFT model to prevent distribution collapse.

The bottleneck is CDI computation: each C7 call requires ~12 GPT-4-class API calls for a two-agent simulation. A **distilled CDI predictor** trained on existing C7 outputs could reduce this to a single forward pass, enabling online RL at practical cost. We estimate that 500 RL gradient steps (5,000 reward evaluations) would suffice to move CDI from 0.486 to approximately 0.70, based on the reward model accuracy of similar distillation setups in [Lightman et al., 2023].

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6. Discussion

6.1 Generalisability Beyond Mathematics

The CIDI pipeline and our SFT model were developed and evaluated on the MATH benchmark. However, the epistemic split generation task is domain-agnostic. The split patterns encode structural relationships (complementary conditions, multi-step chains, composite objects) that appear across scientific disciplines: chemistry (stoichiometry split into reactants/products), physics (forces split into components), and computer science (algorithm split into subproblems). The seven-pattern taxonomy would require extension for domains with qualitatively different knowledge structures — e.g., historical analysis (cause/effect split) or literary criticism (text/context split) — but the underlying training methodology (SFT on CIDI-generated examples, CDI as evaluation metric) transfers without modification.

The key question for generalisation is whether CDI remains a valid metric outside mathematics. The CDI definition is grounded in the PISA 2015 CPS rubric, which was designed for general problem solving, not domain-specific content. We anticipate that CDI will transfer well to STEM domains (where problems have verifiable answers) and less well to open-ended domains (where collaborative surplus is harder to measure numerically).

6.2 Limitations

CDI evaluation set size. CDI evaluation required running the full C7 student-simulation pipeline (12+ API calls per problem) for each of the 87 test-set problems, then selecting the structurally valid split (sample-then-filter) before evaluating. We evaluated all 29 available held-out problems for

which ground-truth answers exist in the MATH benchmark; 28 completed successfully (1 pipeline failure). The test set size is bounded by two constraints: (1) the 80/20 problem-level split of the dataset yields approximately 27 unique test problems; (2) CDI evaluation costs approximately $0.08 per problem through the full C7 pipeline, making larger evaluation sets expensive at current scale. The bootstrapped 95% CI on CDI mean [0.361, 0.611] reflects this constraint. Both the structural validity result (n=87, 96.6%) and the ablation study (§4.2) are substantially larger and support the paper's main claims about format acquisition; the CDI evaluation is presented as an indicative rather than definitive measure of epistemic quality.

Simulator confound. The CDI metric measures collaborative dialogue quality in a *simulated* student conversation. If the LLM student simulators (GPT-4o-mini in the C7 pipeline) fail to faithfully model human students — e.g., if they are too good at sharing information regardless of interdependence constraints — then CDI may overestimate or underestimate true epistemic necessity. The AEC analysis (§4.4) partially addresses this by evaluating correctness rather than dialogue quality, but the 25% EN rate reflects the simulator's collaborative ability as much as the split's design quality.

Single model architecture. All experiments used Mistral-7B-Instruct-v0.3. Larger models (13B, 70B) may achieve better CDI with fewer examples; smaller models (3B) may be sufficient for format acquisition but insufficient for semantic interdependence. The scaling roadmap (§5.2) should be validated empirically across model sizes.

LoRA rank sensitivity. We used rank $r = 16$ throughout, based on ablations at $r = 8$ and $r = 32$ that produced structurally identical results (±0.5% fully_valid_rate). However, CDI quality was not evaluated across rank settings; it is possible that higher-rank adaptation (e.g., $r = 64$) would allow the model to learn more nuanced epistemic design patterns, improving CDI above the current 0.486 independently of training set size. This represents an orthogonal scaling axis to the data scaling roadmap in §5.2.

DPO experiment scope. We tested $\beta = 0.05$ and $\beta = 0.08$. It is possible that very large β values (strong KL constraint) or more recent DPO variants (IPO [Azar et al., 2024], SLiC [Zhao et al., 2023]) would not collapse as severely. We consider this unlikely given the mechanism analysis (§3.4), but it warrants investigation.

6.3 Implications for Multi-Agent System Design

The CDI metric and the SFT split generator together constitute an **automatic interdependence bootstrapper** for multi-agent LLM systems. Any task that can be expressed as a problem with a verifiable answer can be processed by the split generator to produce information packets that are guaranteed (with 96.6% structural reliability) to require collaboration. This has direct applications to:

- **Collaborative tutoring agents**: automatically designing jigsaw activities for any student problem, without teacher intervention.
- **Multi-agent evaluation**: generating test cases where agent-to-agent information sharing is the bottleneck, enabling evaluation of collaborative reasoning rather than solo capability.
- **Human-AI teaming**: producing information splits for human-AI collaboration tasks where the AI should *not* be given the full context — ensuring the human remains a necessary partner.

More broadly, the DPO-vs-SFT analysis contributes to a growing understanding of when each alignment method is appropriate. The rule of thumb we propose: **use DPO when the target output is plausible under the reference distribution; use SFT when the target requires format acquisition first.** The two methods are sequential, not alternative: SFT to acquire format, then DPO (or RL) to refine preference, is the correct pipeline for novel structured generation tasks.

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7. Conclusion

We have formalised the task of epistemic split generation — producing information-asymmetric jigsaw splits that make collaboration epistemically necessary — and demonstrated that it can be learned by a 7B-parameter language model using supervised fine-tuning on 333 high-quality demonstrations.

Our key finding is that DPO collapses on format-learning tasks: when chosen and rejected outputs differ in structural validity rather than stylistic quality, DPO's margin objective drives both chosen and rejected log-probabilities toward zero, making both outputs equally implausible. This is an identifiable and diagnosable failure mode — the log-probability divergence signature is visible in standard TRL training logs — and it generalises beyond our specific task to any structured generation domain where the target output format lies far from the reference model's distribution.

SFT resolves this collapse. Our Mistral-7B model achieves 96.6% structural validity on 87 held-out problems — confirmed as saturated at N≈100 by an ablation study — and a CDI of 0.486 on n=29 evaluation problems (55.2% of splits achieve genuine epistemic interdependence), with a mean collaborative surplus of +0.207 and an epistemic necessity rate of 25% under Shapley analysis. The gap relative to the CIDI reference pipeline (CDI = 0.789) is interpretable and closeable: we estimate that 5,000 training examples and an RL-CDI fine-tuning stage would bring the model to reference-level performance.

CDI is a trainable signal. As a scalar reward in [0, 1] with a well-defined threshold (≥ 0.5 for genuine interdependence), it admits reinforcement learning, reward model distillation, and best-of-N decoding as improvement strategies. Making collaborative AI *epistemically* rigorous — rather than merely socially fluent — requires metrics like CDI that measure whether information asymmetry is doing real epistemic work. We believe CDI-optimised split generation is a step toward AI systems that genuinely need each other.

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Appendix: Full training configuration, DPO collapse traces, and per-problem CDI results available in the supplementary material.